APPENDIX 1

MARKING RUBRICS

Component Title	Power flow studies using Gauss-Seidel and Newton-Raphson				Percentage (%)		
Criteria	Score and Descriptors					Weight (%)	Marks
	Excellent (5)	Good (4)	Average (3)	Need Improvement (2)	Poor (1)		
System design	100% reliable and unique isolated network topology design and using logical parameters of industry data	80% reliable and unique isolated network topology design and using logical parameters of industry data	60% reliable and unique isolated network topology design and using logical parameters of industry data	40% reliable and unique isolated network topology design and using logical parameters of industry data	plagiarized isolated network topology design	2	
Manual calculation to solve the voltages using Gauss-Seidel method	100% detail steps of calculation shown with comprehensively and correctly evaluated	80% detail steps of calculation shown with comprehensively and correctly evaluated	60% detail steps of calculation shown with comprehensively and correctly evaluated	40% detail steps of calculation shown with comprehensively and correctly evaluated	less than 40% detail steps of calculation shown with comprehensively and correctly evaluated	4	
PowerWorld Simulator simulation using Gauss-Seidel method	100% detail simulation conducted and compared with manual calculation for every iteration	80% detail simulation conducted and compared with manual calculation for every iteration	60% detail simulation conducted and compared with manual calculation for every iteration	40% detail simulation conducted and compared with manual calculation for every iteration	less than 40% detail simulation conducted and compared with manual calculation for every iteration	2	
Manual calculation to solve the voltages using Newton-Raphson method	100% detail steps of calculation shown with comprehensively and correctly evaluated	80% detail steps of calculation shown with comprehensively and correctly evaluated	60% detail steps of calculation shown with comprehensively and correctly evaluated	40% detail steps of calculation shown with comprehensively and correctly evaluated	less than 40% detail steps of calculation shown with comprehensively and correctly evaluated	4	
PowerWorld Simulator	100% detail simulation conducted	80% detail simulation conducted	60% detail simulation conducted	40% detail simulation conducted	less than 40% detail simulation	2	

simulation using Newton-Raphson method	**and compared with manual calculation for every iteration**	**and compared with manual calculation for every iteration**	**and compared with manual calculation for every iteration**	**and compared with manual calculation for every iteration**	**conducted and compared with manual calculation for every iteration**		
Overall power flow results discussion and comparison of Gauss-Seidel and Newton-Raphson methods	**100% detail overall power flow results being discussed with comparison of both Gauss-Seidel and Newton-Raphson methods and justification for method adoption**	**80% detail overall power flow results being discussed with comparison of both Gauss-Seidel and Newton-Raphson methods and justification for method adoption**	**60% detail overall power flow results being discussed with comparison of both Gauss-Seidel and Newton-Raphson methods and justification for method adoption**	**40% detail overall power flow results being discussed with comparison of both Gauss-Seidel and Newton-Raphson methods and justification for method adoption**	**less than 40% detail overall power flow results being discussed with comparison of both Gauss-Seidel and Newton-Raphson methods and justification for method adoption**	**4**	
Create a situation when any generator's reactive power output reaches its limit	**100% detail situation created for generator bus becomes load bus with observations and findings**	**80% detail situation created for generator bus becomes load bus with observations and findings**	**60% detail situation created for generator bus becomes load bus with observations and findings**	**40% detail situation created for generator bus becomes load bus with observations and findings**	**Less than 40% detail situation created for generator bus becomes load bus with observations and findings**	**2**	
					TOTAL		

Note to students: Please include the marking rubric when submitting your coursework.